

Development Environment & SDK

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Developer Technical Services

Agenda

- **System Configuration**
  - Hardware requirements: Workstation & peripherals
  - Ubuntu Installation
- **PDN – Palm Developer Network**
  - Signing up/Logging in
  - Downloading SDK/ROMs
- **SDK**
  - Extraction & installation
- **Documentation**
  - Getting Started Guide, Developer Guide & API Guide
- **SDK layout**
  - Samples directory
  - sysroot & toolroot directories
- **Developer Resources**



System Configuration

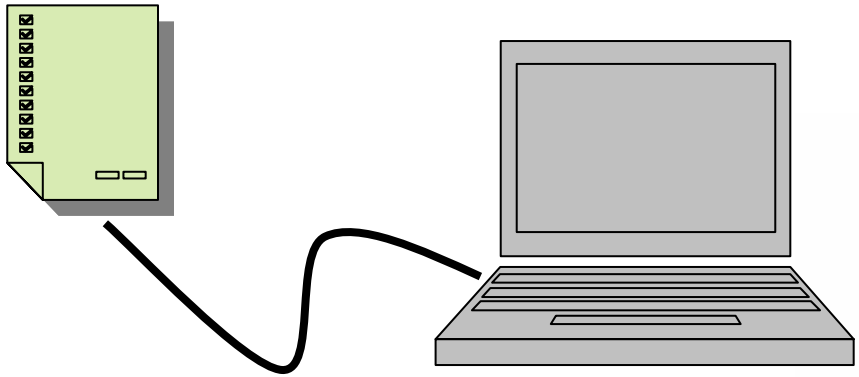
- Linux
  - There are many different distributions. Internally we use Ubuntu and Redhat
  - Use your favorite editor/IDE
    - Eclipse, Emacs, kdevelope
- Must use the included toolchain
  - gcc, ld
  - gdb and gdbserver
- Need root access.
  - Or entry in sudoers file
- Write access to /opt
  - SDK will install toolchain into this directory
- A network card on a private IP network.
  - Necessary for nfs-mounting/remote debugging over Ethernet
- Fusion module which will be installed by SDK
  - Need kernel sources to compile the module against.



System Configuration: SD Card



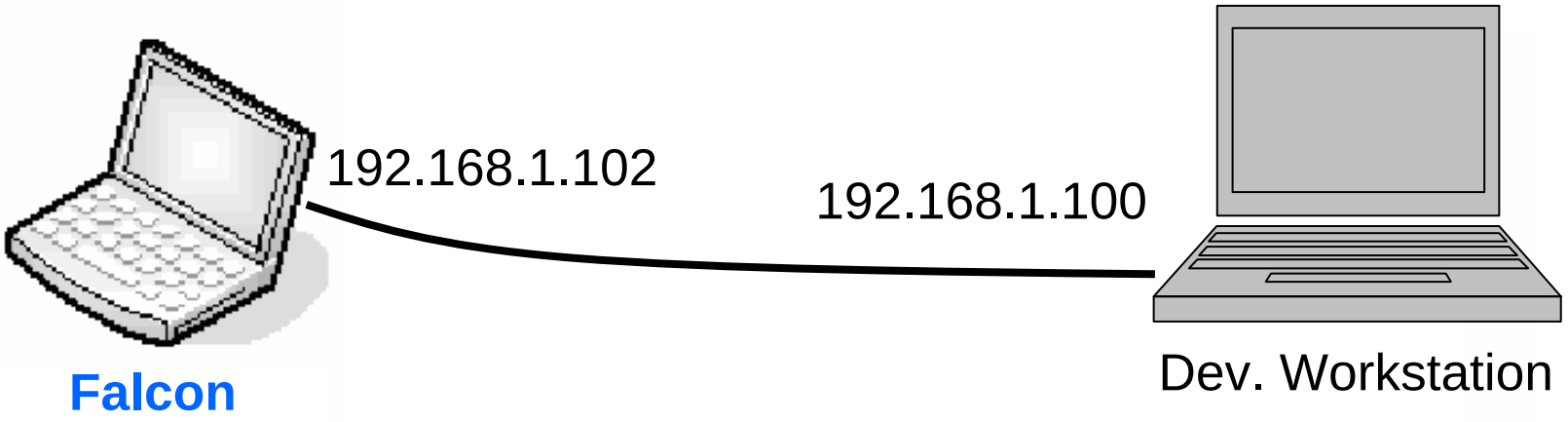
Falcon



Dev. Workstation

- Purchase an SD Card
  - Required to Flash device.
- Enable SD/MTD support in the kernel of the workstation

System Configuration: NFS Setup



- File transfer and Remote Debugging are feasible via private network connection.
  - Add a 2nd Ethernet card to your workstation. (wifi will work)
  - USB-Ethernet Adapter and Crossover Cable facilitate communication on a private network.

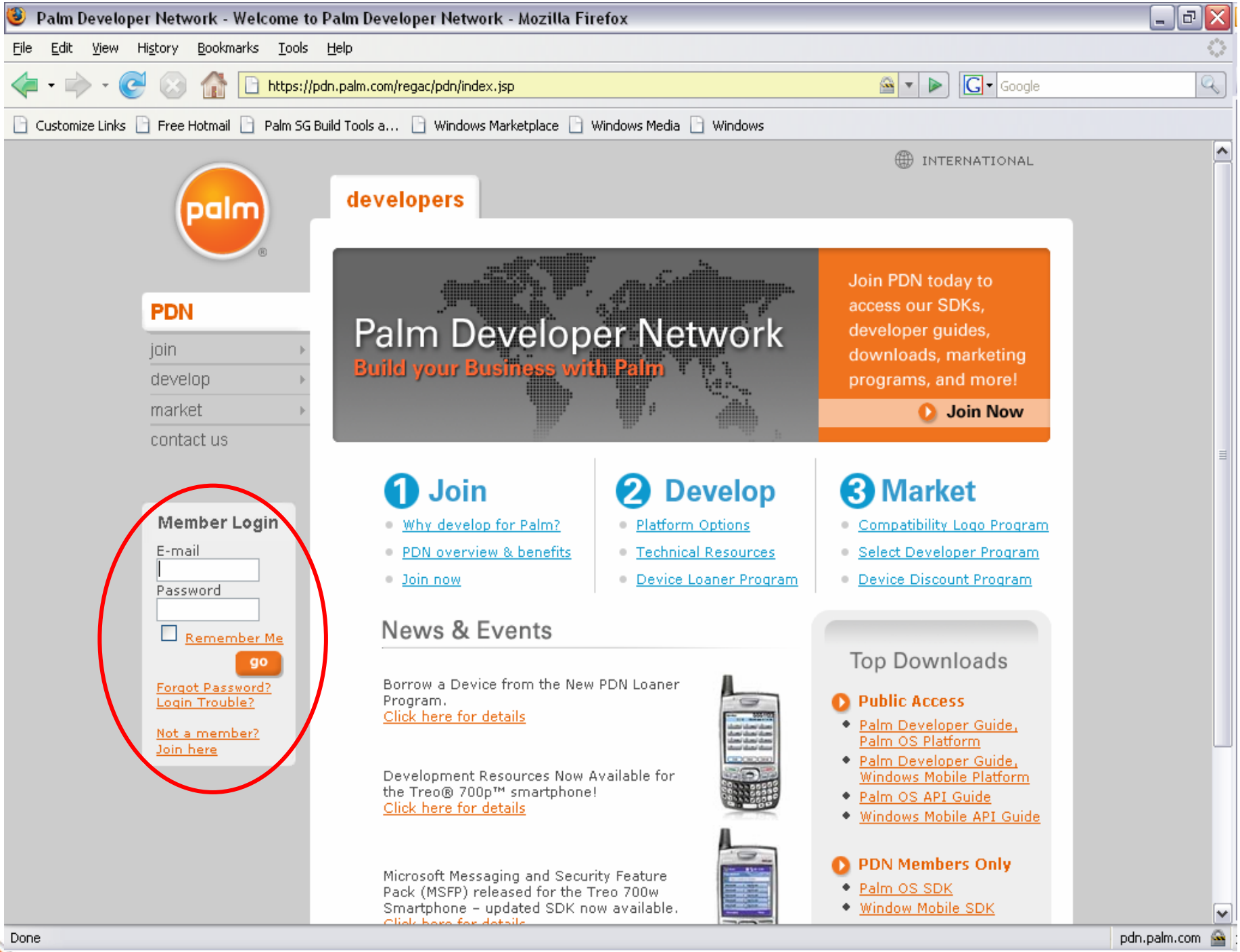
Ubuntu installation

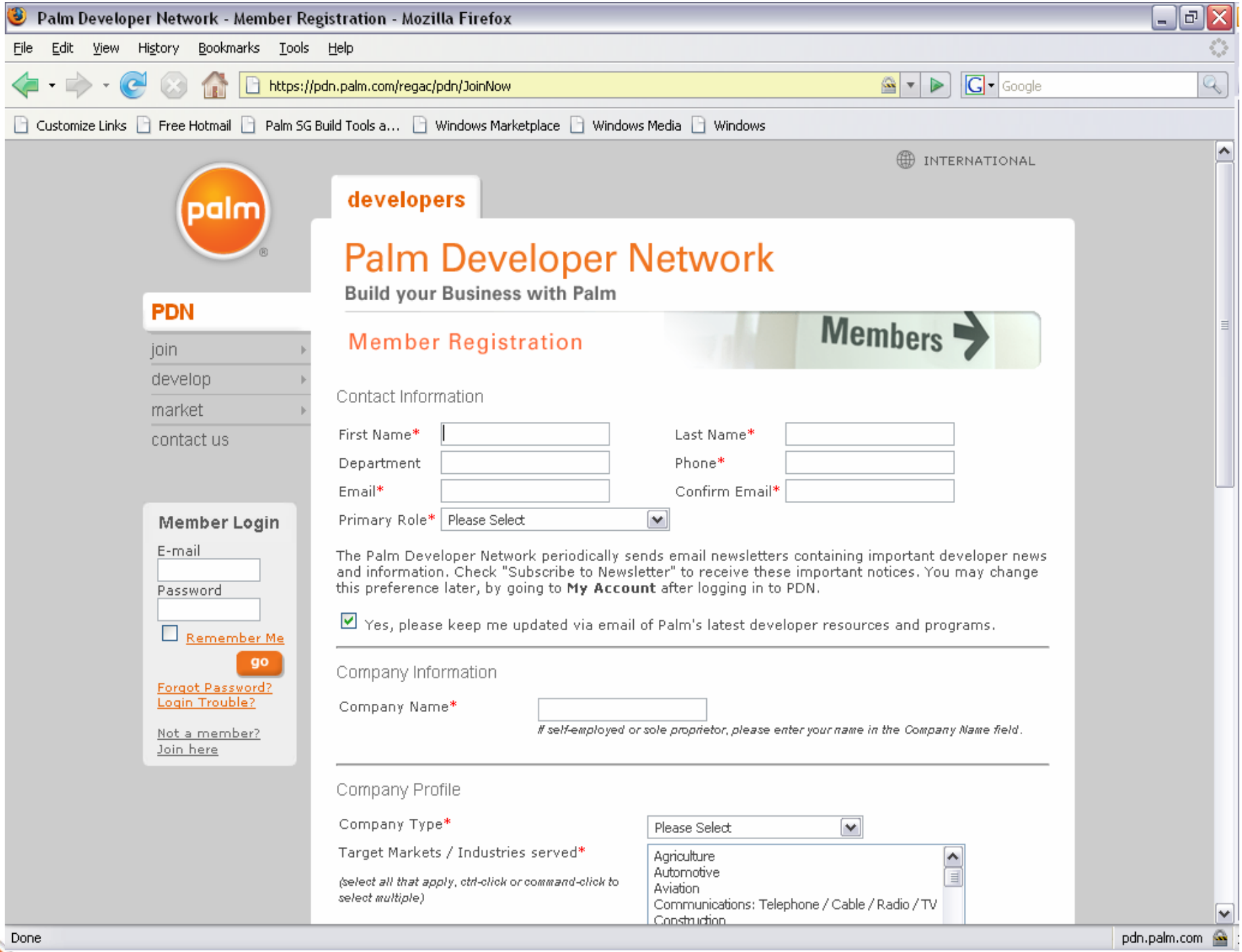
- www.ubuntu.com
  - Download desktop version 6.06 also called “Drapper Drake”
  - Create installation CD
  - Run CD and install Ubuntu
- Updating Ubuntu
  - $ sudo apt-get update
  - Synaptic Package Manager (System->Administration)
- Download Kernel headers – Required to install fusion
  - $ uname –a
  - $ sudo apt-get install build-essential linux-headers-2.6.15-xx-xxx”
- Update list of available packages
  - apt-get update
  - Install xchm package
    - apt-get install xchm
  - Install nfs-user-server
    - apt-get install nfs-user-server



Software Development Kit (SDK)

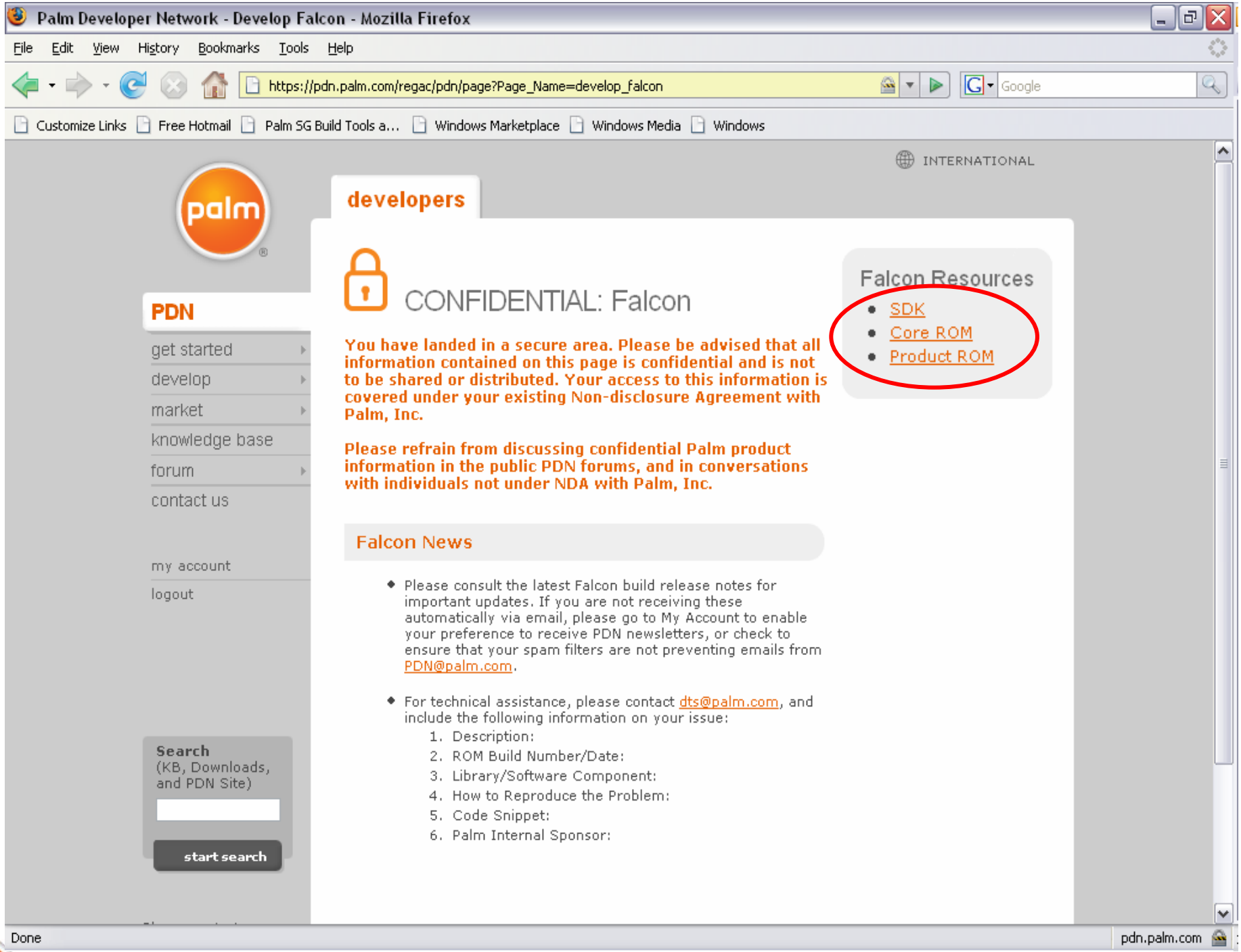


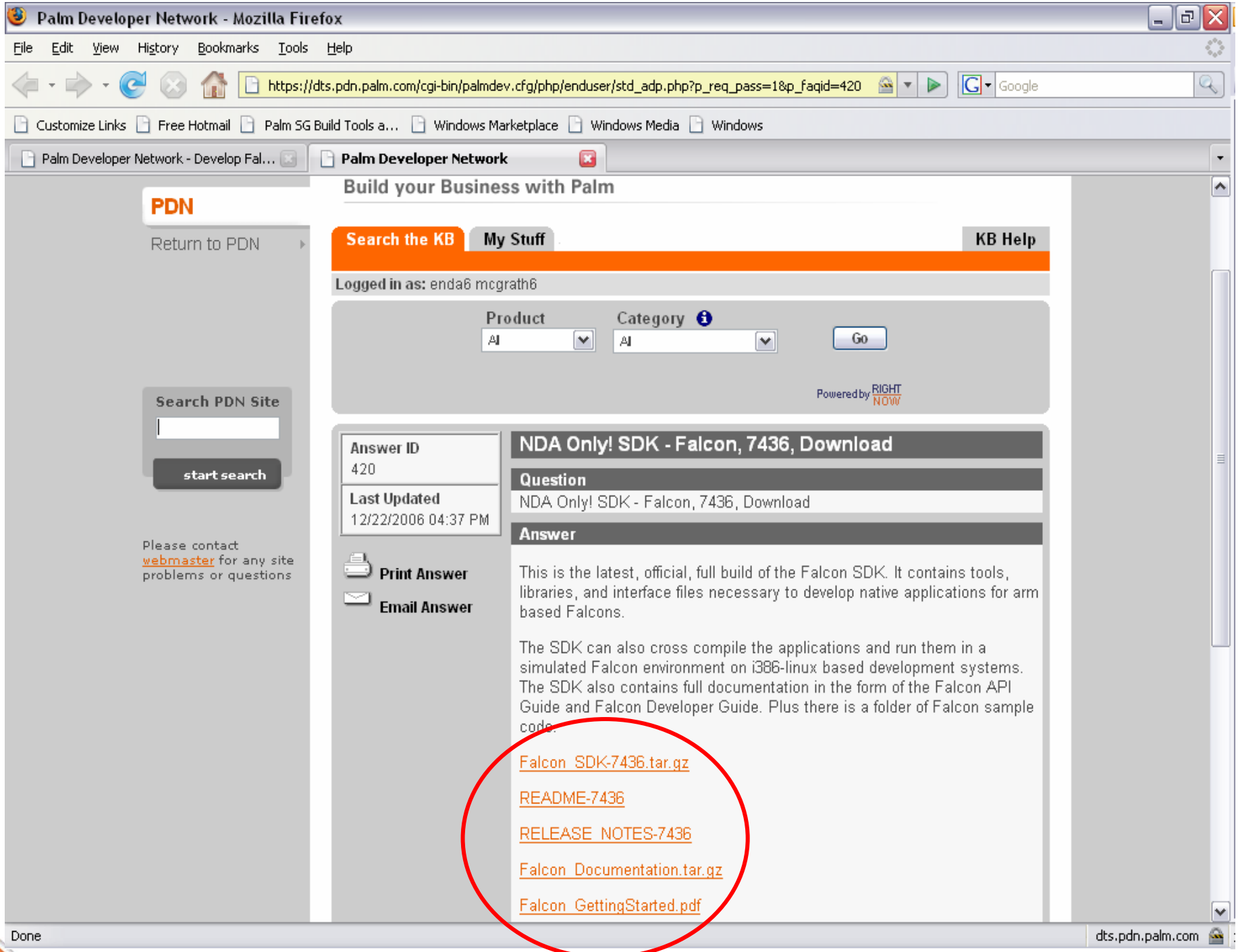

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Why develop for Palm?
PDN overview & benefits
Join now
2 Develop
Platform Options
Technical Resources
Device Loaner Program
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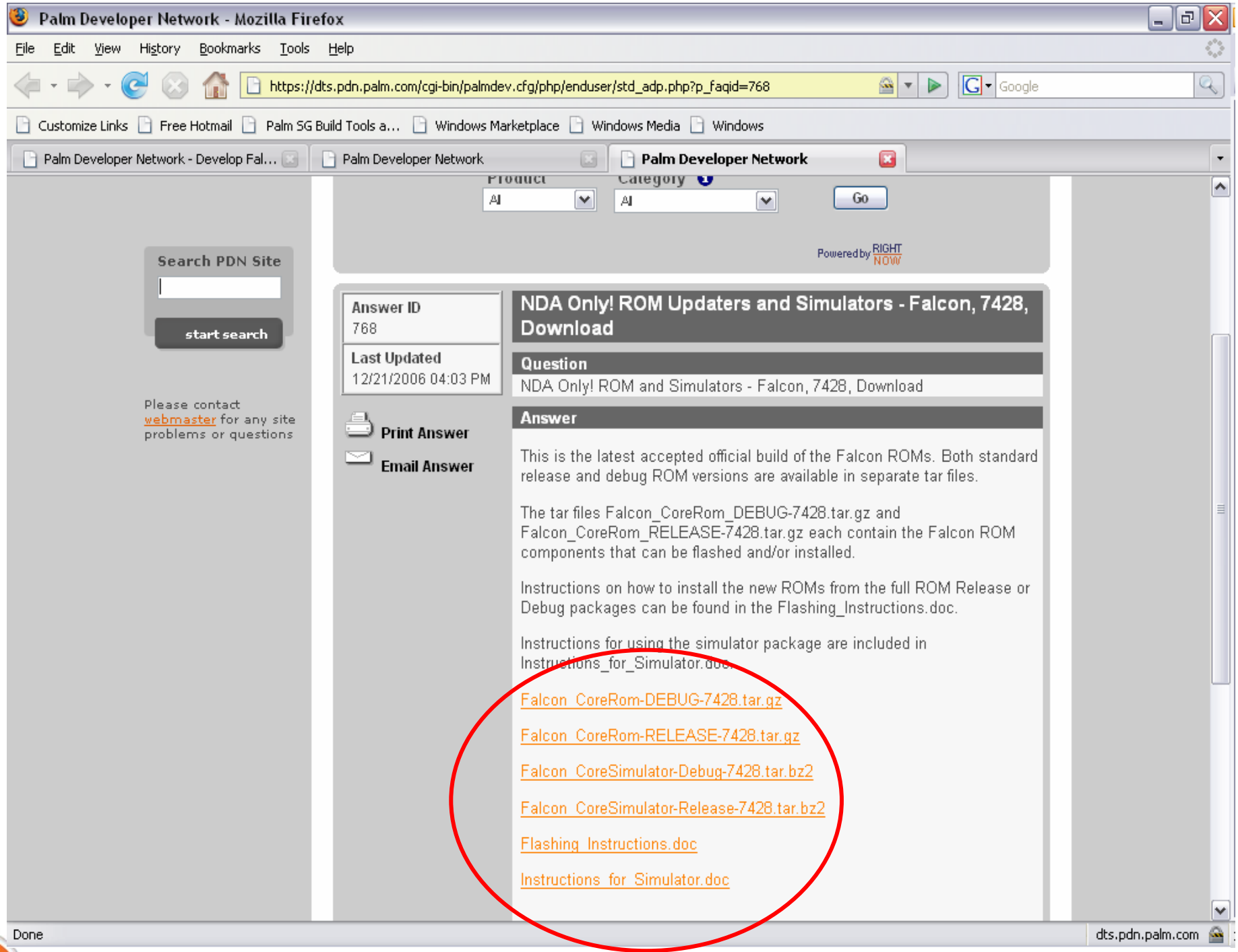

Palm Developer Network - Member Registration - Mozilla Firefox
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Google
Customize Links Free Hotmail Palm SG Build Tools a... Windows Marketplace Windows Media Windows
INTERNATIONAL
developers
Palm Developer Network
Build your Business with Palm
Member Registration
Members
PDN
join
develop
market
contact us
Member Login
E-mail
Password
Remember Me
go
Forgot Password?
Login Trouble?
Not a member?
Join here
Contact Information
First Name*
Last Name*
Department
Phone*
Email*
Confirm Email*
Primary Role*
Please Select
The Palm Developer Network periodically sends email newsletters containing important developer news and information. Check "Subscribe to Newsletter" to receive these important notices. You may change this preference later, by going to My Account after logging in to PDN.
Yes, please keep me updated via email of Palm's latest developer resources and programs.
Company Information
Company Name*
If self-employed or sole proprietor, please enter your name in the Company Name field.
Company Profile
Company Type*
Please Select
Target Markets / Industries served*
(select all that apply, ctrl-click or command-click to select multiple)
Agriculture
Automotive
Aviation
Communications: Telephone / Cable / Radio / TV
Construction
Done
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Accessing secure Falcon site


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You have landed in a secure area. Please be advised that all information contained on this page is confidential and is not to be shared or distributed. Your access to this information is covered under your existing Non-disclosure Agreement with Palm, Inc.
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Falcon Resources
SDK
Core ROM
Product ROM
Falcon News
Please consult the latest Falcon build release notes for important updates. If you are not receiving these automatically via email, please go to My Account to enable your preference to receive PDN newsletters, or check to ensure that your spam filters are not preventing emails from PDN@palm.com.
For technical assistance, please contact dts@palm.com, and include the following information on your issue:
1. Description:
2. ROM Build Number/Date:
3. Library/Software Component:
4. How to Reproduce the Problem:
5. Code Snippet:
6. Palm Internal Sponsor:
Done
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Product
Category
All
All
Go
Powered by RIGHT NOW
Search PDN Site
start search
Please contact webmaster for any site problems or questions
Answer ID
420
Last Updated
12/22/2006 04:37 PM
Print Answer
Email Answer
NDA Only! SDK - Falcon, 7436, Download
Question
NDA Only! SDK - Falcon, 7436, Download
Answer
This is the latest, official, full build of the Falcon SDK. It contains tools, libraries, and interface files necessary to develop native applications for arm based Falcons.
The SDK can also cross compile the applications and run them in a simulated Falcon environment on i386-linux based development systems. The SDK also contains full documentation in the form of the Falcon API Guide and Falcon Developer Guide. Plus there is a folder of Falcon sample code.
Falcon_SDK-7436.tar.gz
README-7436
RELEASE_NOTES-7436
Falcon_Documentation.tar.gz
Falcon_GettingStarted.pdf
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Palm Developer Network - Develop Fal...
Palm Developer Network
Palm Developer Network
Product
All
Category
All
Go
Powered by RIGHT NOW
Search PDN Site
start search
Please contact webmaster for any site problems or questions
Answer ID
768
Last Updated
12/21/2006 04:03 PM
Print Answer
Email Answer
NDA Only! ROM Updaters and Simulators - Falcon, 7428, Download
Question
NDA Only! ROM and Simulators - Falcon, 7428, Download
Answer
This is the latest accepted official build of the Falcon ROMs. Both standard release and debug ROM versions are available in separate tar files.
The tar files Falcon_CoreRom_DEBUG-7428.tar.gz and Falcon_CoreRom_RELEASE-7428.tar.gz each contain the Falcon ROM components that can be flashed and/or installed.
Instructions on how to install the new ROMs from the full ROM Release or Debug packages can be found in the Flashing_Instructions.doc.
Instructions for using the simulator package are included in Instructions_for_Simulator.doc.
Falcon_CoreRom-DEBUG-7428.tar.gz
Falcon_CoreRom-RELEASE-7428.tar.gz
Falcon_CoreSimulator-Debug-7428.tar.bz2
Falcon_CoreSimulator-Release-7428.tar.bz2
Flashing_Instructions.doc
Instructions_for_Simulator.doc
Done
dts.pdn.palm.com

PDN Welcome doc

- **Login to PDN**
- **Navigate to Knowledge Base**
- **Lookup answer 818**
- **Or use the following link**

  https://dts.pdn.palm.com/cgi-bin/palmdev.cfg/php/enduser/std_adp.php?p_faqid=818



PDN Downloads

- **Falcon SDK**
  - Contains the toolchain, system root for building simulator and ARM applications, documentation. Check Release Notes
- **Falcon Core ROM**
  - Falcon_CoreRom-RELEASE-xxxx.tar.gz contains the files necessary to flash a device with the Core Rom images. Contains the simulator package for the Core Image
- **Falcon Product ROM**
  - Falcon Product ROM Image contains the Falcon Core with 3rd party applications. Contains the simulator package for the Product Image

- ROM and SDK tar.gz packages must match
- Select the Apps Key, scroll to System Preferences, select System Info. Build number will be listed in the output.

Extracting the SDK

```
emcgrath@emcgrath-laptop:~/Falcon$ ls
Build                                          Falcon_SDK-7436.tar.gz
BUILD_VERSION                                  Flashing_Instructions.doc
Falcon_CoreRom-RELEASE-7428.tar.gz             Instructions_for_Simulator.doc
Falcon_CoreSimulator-Release-7428.tar.bz2      README-7436.txt
Falcon_Documentation.tar.gz                    RELEASE_NOTES-7436.txt
Falcon_GettingStarted.pdf                      rootfs.tar.gz
```



Extracting the SDK

```
emcgrath@emcgrath:~/Falcon$ tar -xvf Falcon_SDK-7457.tar.gz
Build/
Build/99-fusion.rules
Build/docs.tar.bz2
Build/EclipseHollySdkPlugin.zip
Build/Eclipse_Holly_Sdk_Plugin_ReadMe.txt
Build/i386-linux-fusion.tar.bz2
Build/README
Build/RELEASE_NOTES
Build/samples.tar.bz2
Build/sdk-installer.sh
Build/sdk_arm-debug-sysroot.tar.bz2
Build/sdk_arm-sysroot.tar.bz2
Build/sdk_i386-sysroot.tar.bz2
Build/sysroot-arm.tar.bz2
Build/sysroot-i386.tar.bz2
Build/toolroot-arm.tar.bz2
Build/toolroot-i386.tar.bz2
Build/update-sysroot.sh
emcgrath@emcgrath:~/Falcon$
```



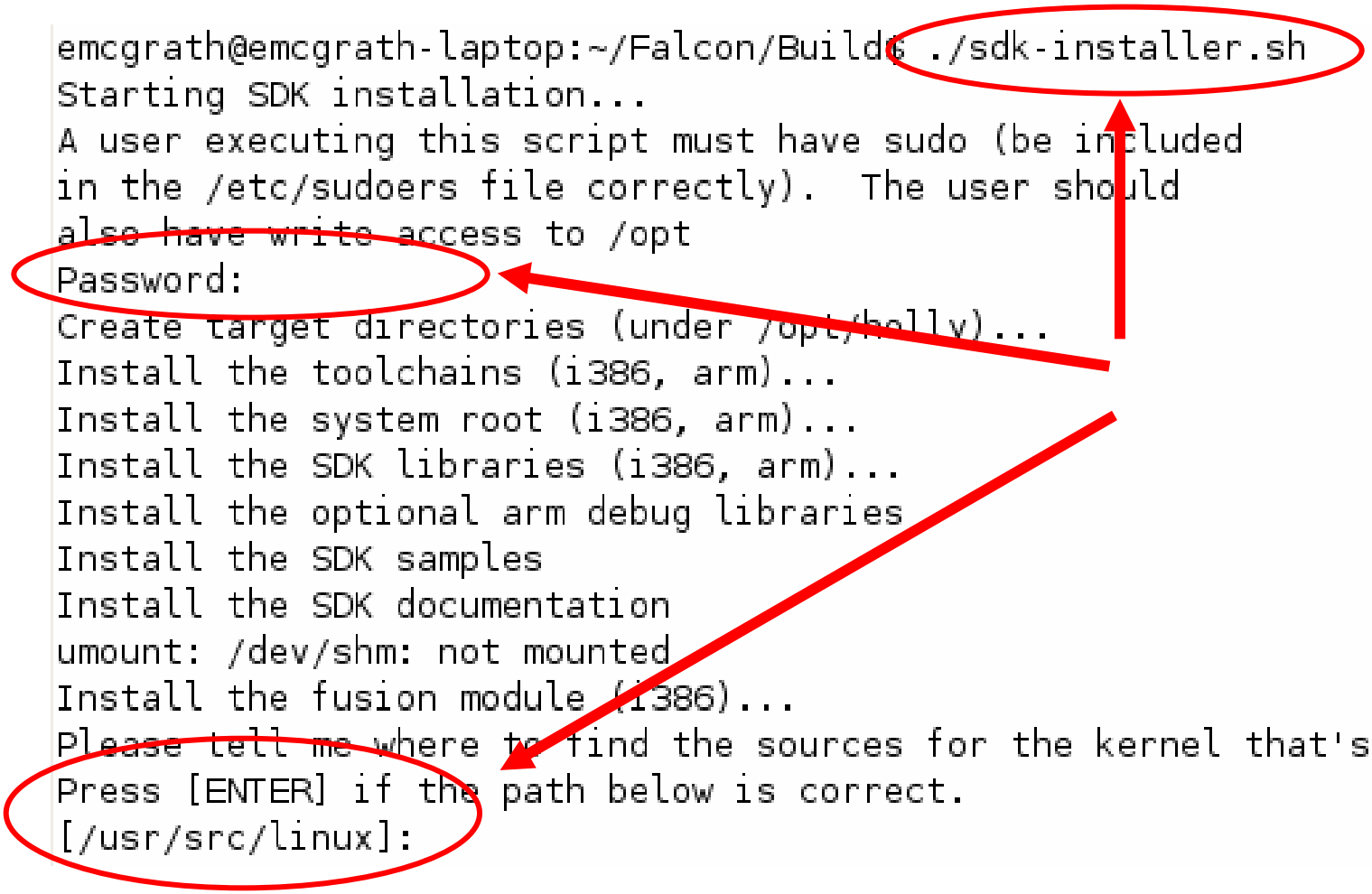
Ubuntu config before SDK installation

- **Set the permissions on the fusion dev entries**
  - Replace 99-fusion.rules
    - $cd /opt/holly/samples/docs
    - $sudo cp 99-fusion.rules /etc/udev/rules.d/
- **Setting directory permission**
  - $ sudo chmod 644 /etc/udev/rules.d/99-fusion.rules
  - $ sudo chmod 777 /opt



Installing SDK

```
emcgrath@emcgrath-laptop:~/Falcon/Builds ./sdk-installer.sh
Starting SDK installation...
A user executing this script must have sudo (be included
in the /etc/sudoers file correctly).  The user should
also have write access to /opt
Password:
Create target directories (under /opt/holly)...
Install the toolchains (i386, arm)...
Install the system root (i386, arm)...
Install the SDK libraries (i386, arm)...
Install the optional arm debug libraries
Install the SDK samples
Install the SDK documentation
umount: /dev/shm: not mounted
Install the fusion module (i386)...
Please tell me where to find the sources for the kernel that's running.
Press [ENTER] if the path below is correct.
[/usr/src/linux]:
```



Installing SDK

```
install -m 644 linux/include/linux/fusion.h /usr/include/linux
install -d /lib/modules/2.6.15-26-386/drivers/char/fusion
install -m 644 linux/drivers/char/fusion/fusion.ko /lib/modules/2.6.15-26-386/drivers/char/fusion
rm -f /lib/modules/2.6.15-26-386/fusion.ko
/sbin/depmod -ae
make: Leaving directory `/opt/holly/sysroot/i386-linux/src/i386-linux-fusion'
Maguffin SDK requires fusion module to run simulator.
Either manually "modprobe fusion" or add it to startup script.


===> ./fusion-installer.sh successful!

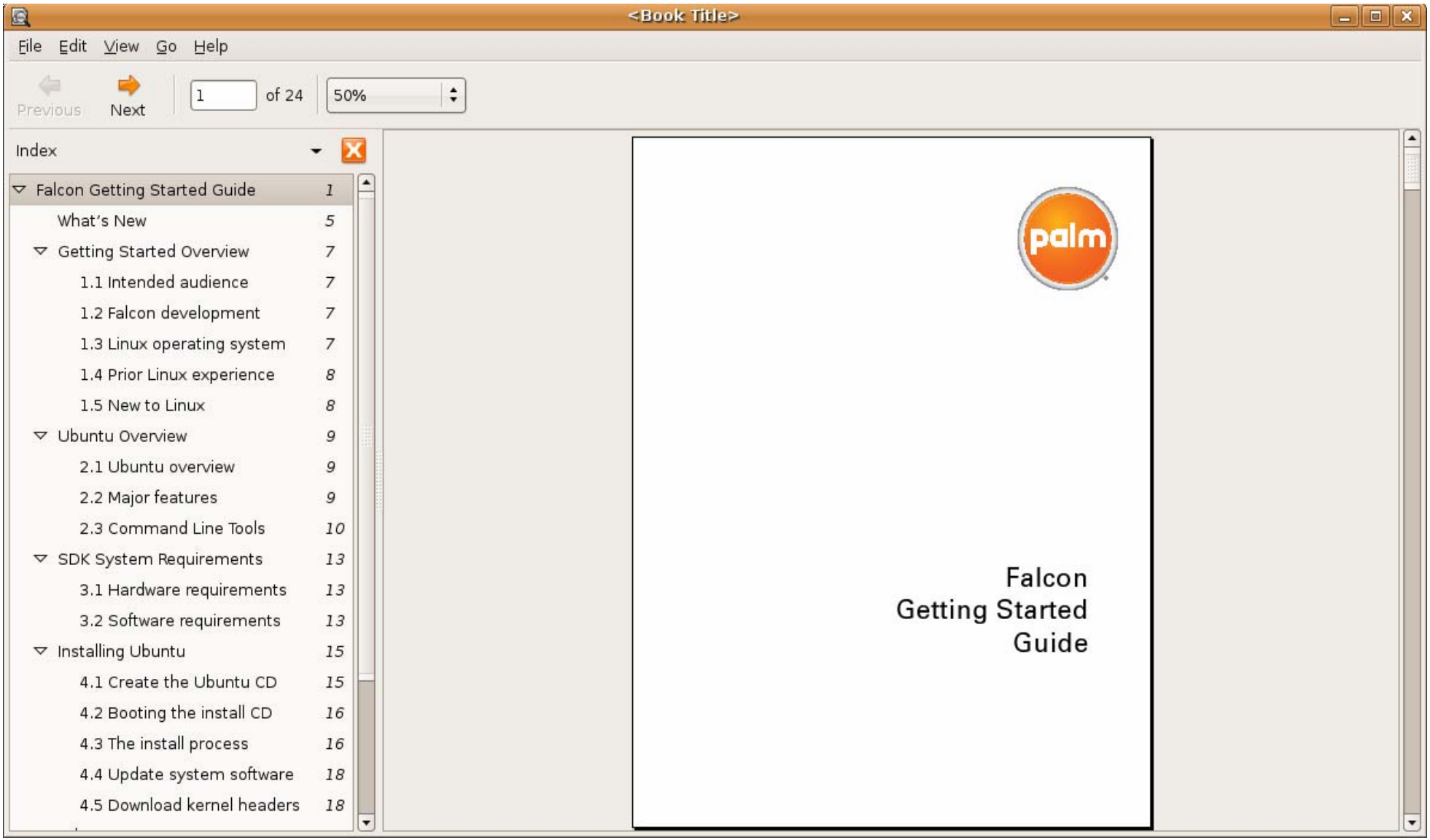

SDK installation done!
emcgrath@emcgrath-laptop:~/Falcon/Build$
```

Getting Started - Documentation

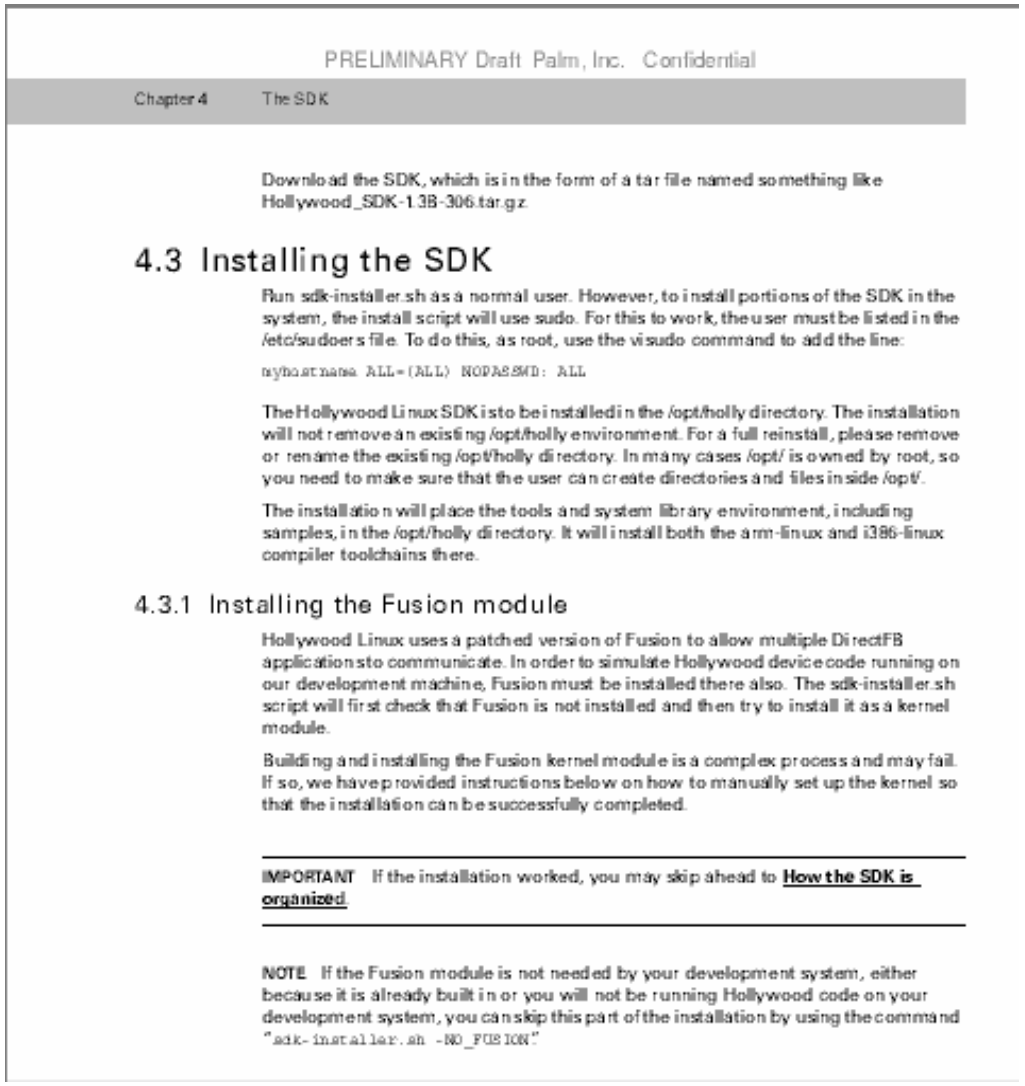
- **Documentation gets installed to the directory;**
  - /opt/holly/docs
    - Getting Started Guide
    - Developer Guide
    - FAQ's
    - Api Guide
      - CHM
      - HTML
    - Configuration files
      - Bash_functions



Falcon Getting Started Guide



SDK: Developer Guide



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Chapter 4 The SDK

Download the SDK, which is in the form of a tar file named something like Hollywood_SDK-1.3B-306.tar.gz.

4.3 Installing the SDK

Run sdk-installer.sh as a normal user. However, to install portions of the SDK in the system, the install script will use sudo. For this to work, the user must be listed in the /etc/sudoers file. To do this, as root, use the visudo command to add the line:

```
myhostname ALL=(ALL) NOPASSWD: ALL
```

The Hollywood Linux SDK is to be installed in the /opt/holly directory. The installation will not remove an existing /opt/holly environment. For a full reinstall, please remove or rename the existing /opt/holly directory. In many cases /opt/ is owned by root, so you need to make sure that the user can create directories and files inside /opt/.

The installation will place the tools and system library environment, including samples, in the /opt/holly directory. It will install both the arm-linux and i386-linux compiler toolchains there.

4.3.1 Installing the Fusion module

Hollywood Linux uses a patched version of Fusion to allow multiple DirectFB applications to communicate. In order to simulate Hollywood device code running on our development machine, Fusion must be installed there also. The sdk-installer.sh script will first check that Fusion is not installed and then try to install it as a kernel module.

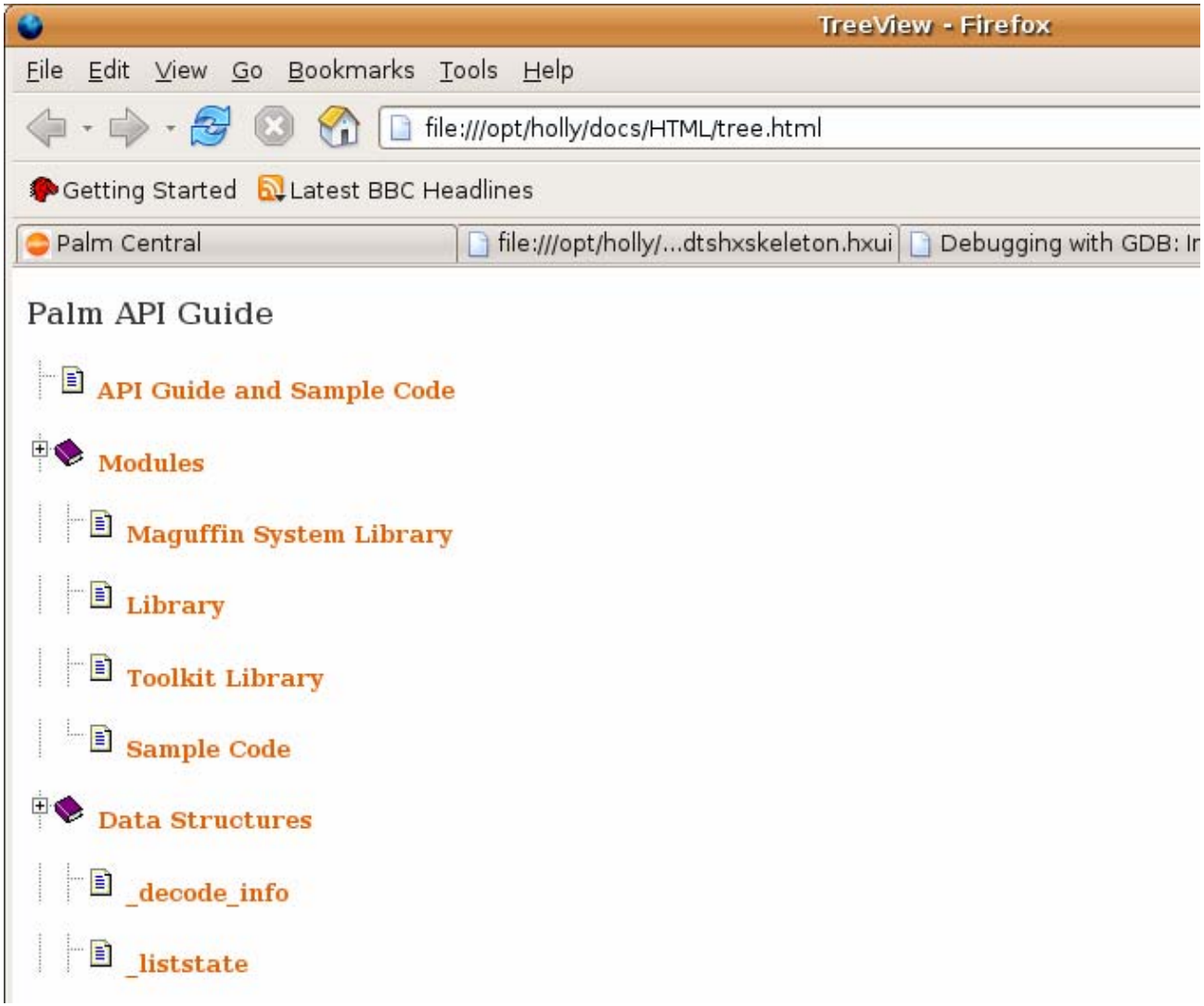
Building and installing the Fusion kernel module is a complex process and may fail. If so, we have provided instructions below on how to manually set up the kernel so that the installation can be successfully completed.

IMPORTANT If the installation worked, you may skip ahead to **How the SDK is organized**.

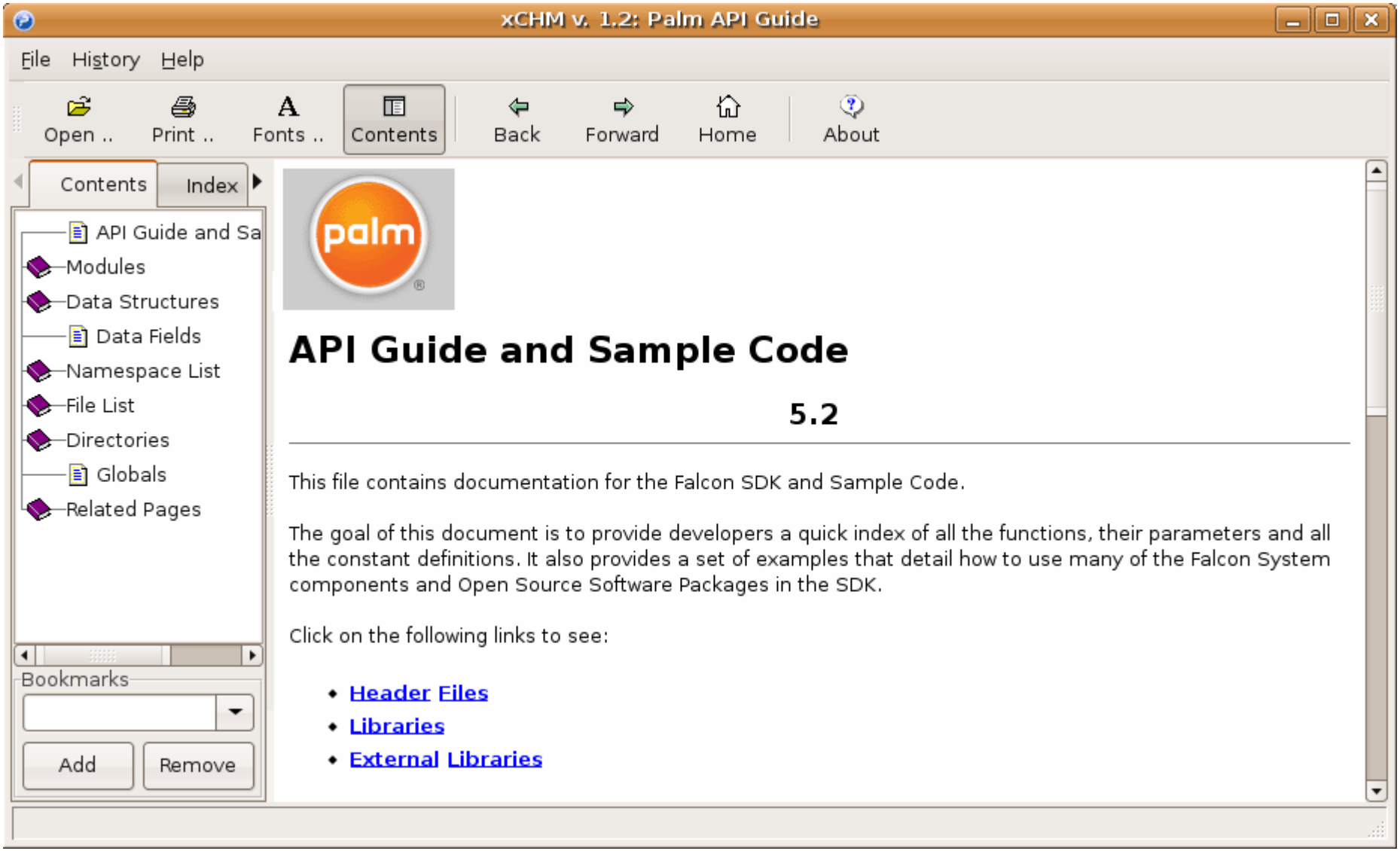
NOTE If the Fusion module is not needed by your development system, either because it is already built in or you will not be running Hollywood code on your development system, you can skip this part of the installation by using the command "`sdk-installer.sh -NO_FUSION`".

SDK: API Guide - /opt/holly/docs/HTML/tree.html

- Add to Bookmarks



SDK: Palm API Guide - CHM



Demo SDK installation

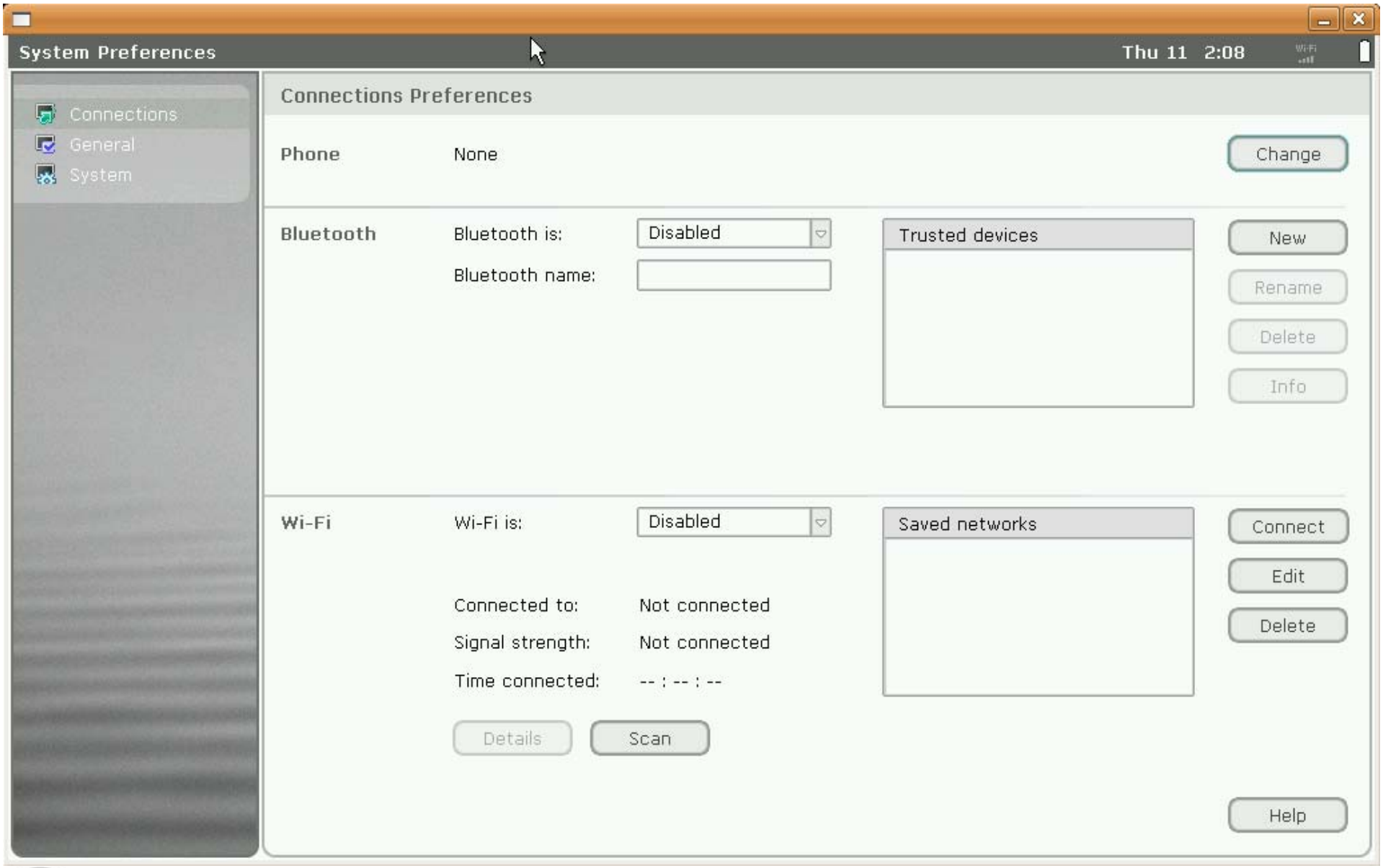


After installation

- **In a terminal navigate to /opt/holly/docs**
  - Falcon_GettingStarted.pdf
  - Falcon_Developer_Guide.pdf
  - Falcon_FAQ.html
  - HTML/API Guide
- **Add functions to .bashrc**
  - $ cat bash_functions >> ~/.bashrc
- **Run Simulator**
  - $ source ~/.bashrc
  - $ dohxtek
    - Simulator Location
      - /opt/holly/sysroot/i386-linux/usr/bin/hxtek



Hxtek - Falcon Simulator



Samples Directory

Sample Name	API feature demonstrated
dtshxskeleton	Starting template, menu's, command listener, resources about dialog
dtshxdffire	Low level directFB calls to provide smooth animation
dtshxpngtoheader	demonstrates the use of directfb-csource utility to read PNG
dtshxquote	Uses webservices to display stock quote, UI API
dtshxrss	Uses the HtmlVIew Container Object to embed HTML. Uses expat library to parse an RSS feed.
dtshxpop	Uses Multiple Containers, ExtendedList, sophisticated Compose dialog, libspopc to retrieve POP3 from yahoo
dtshxtestpattern	Graphics, example
dtshxcalculator	Simple calculator, demonstrates menu, buttons listeners



Compile and run dtshxhello sample

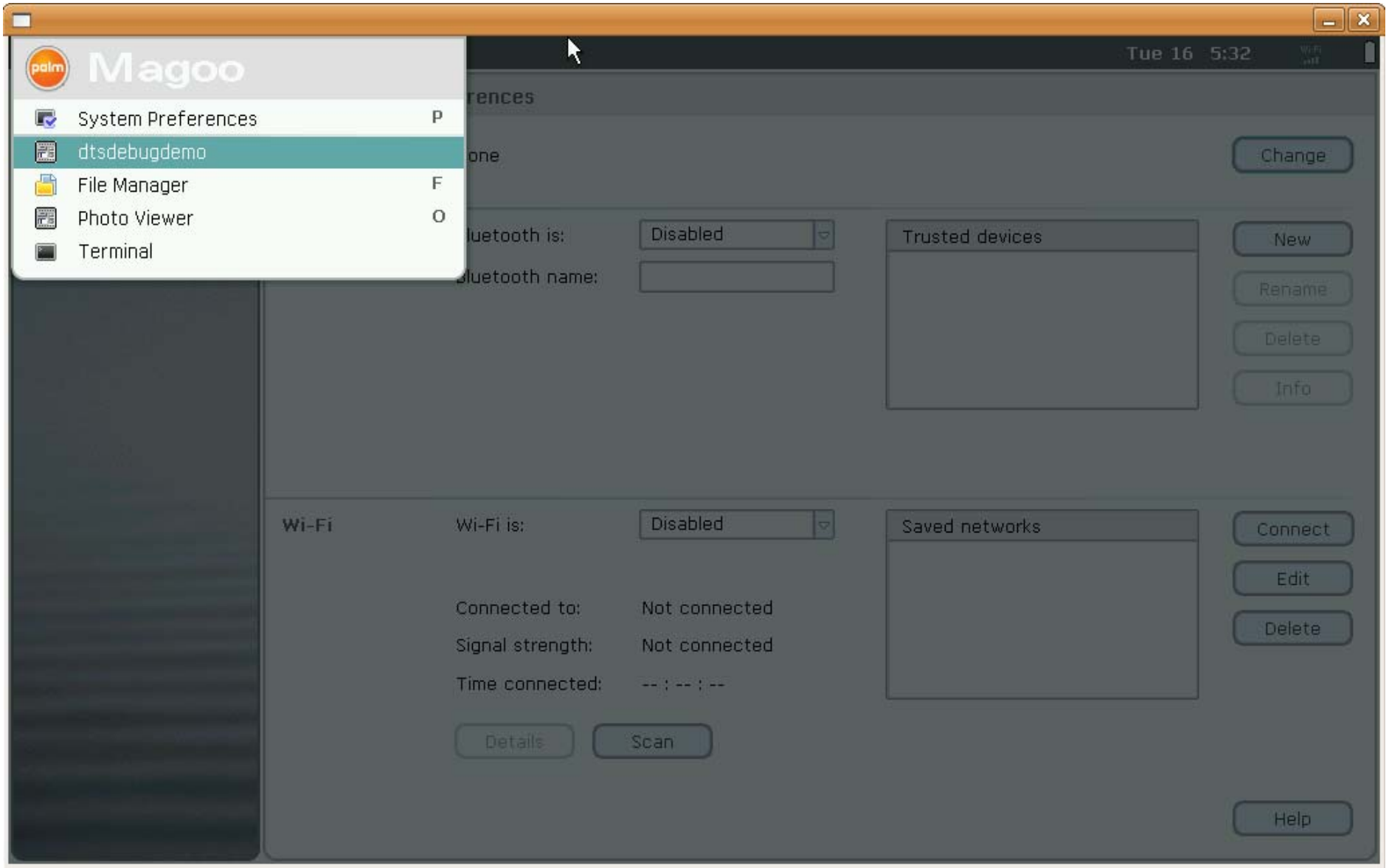
- Navigate to dtshxhello directory
  - $ cd /opt/holly/samples/dtshxhello
- Compile for i386
  - $ make
- Run simulator – using function from bash_functions
  - With application running
    - $ exe dtshxhello
  - With hxtek (application manager) running
    - $ dohxtek



Falcon Simulator – exe hxhello

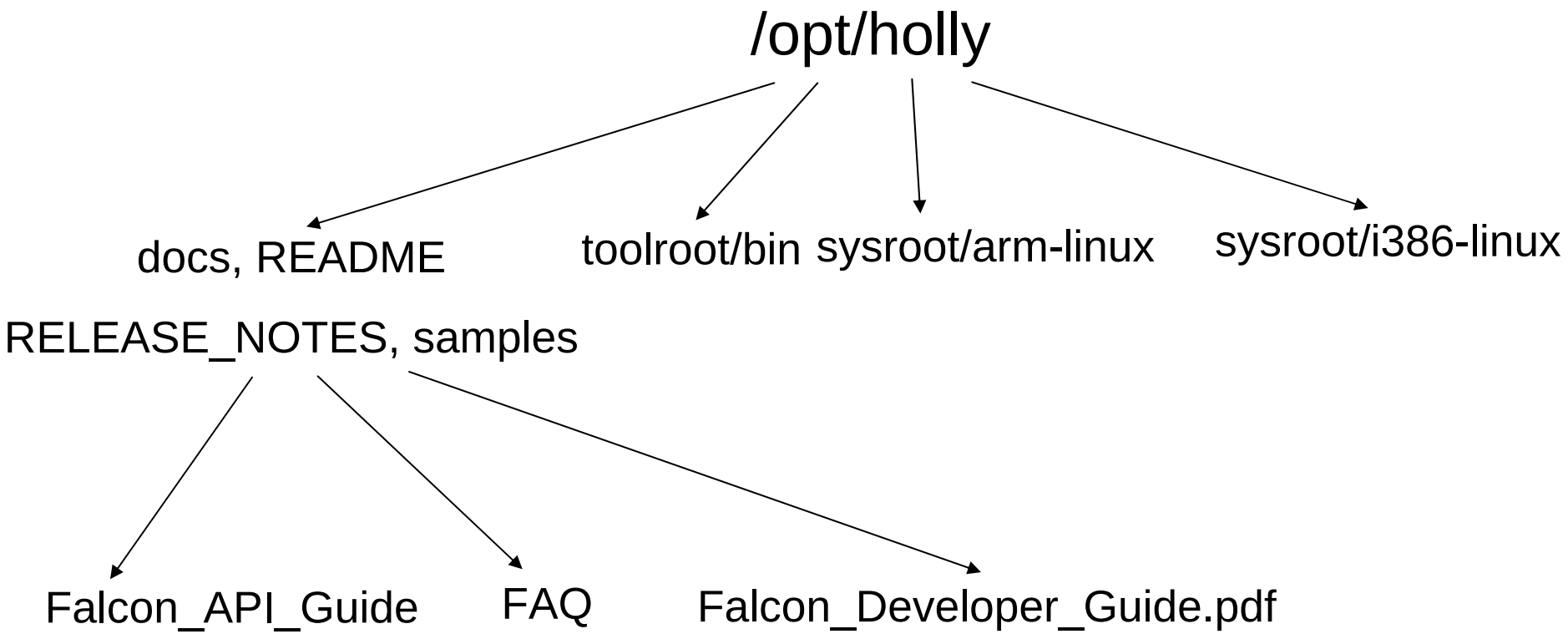


Falcon Simulator – dohxtek

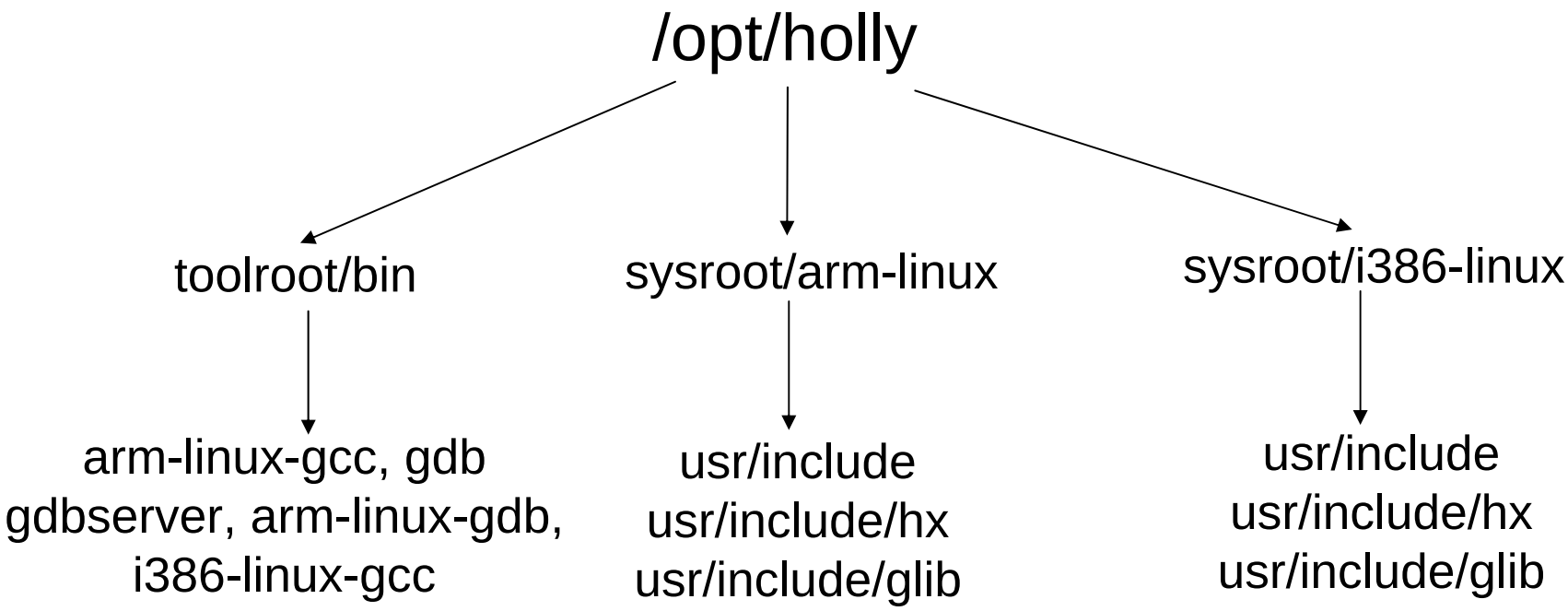


Demo Sample

SDK: Layout


/opt/holly
docs, README
RELEASE_NOTES, samples
toolroot/bin
sysroot/arm-linux
sysroot/i386-linux
Falcon_API_Guide
FAQ
Falcon_Developer_Guide.pdf

SDK: Layout


/opt/holly
toolroot/bin
sysroot/arm-linux
sysroot/i386-linux
arm-linux-gcc, gdb
gdbserver, arm-linux-gdb,
i386-linux-gcc
usr/include
usr/include/hx
usr/include/glib
usr/include
usr/include/hx
usr/include/glib

Developer Resources Overview

- Periodic Updates
  - Email notifying of a new Falcon build posted to PDN
- Front Line Support
  - SDK documentation, developer guide, API guide, FAQ's
  - PDN site
  - **Knowledge Base**
  - Case Management
  - Private Forum
    - Forum->forum postings->Falcon launch developer forum
  - Developer Technical Support (DTS) – devsupport@palm.com
    - Same system as for Palm OS and WinMo
- Verify issue against latest build before reporting
    - Include core dump if possible

